

UNIVERSIDAD POLITÉCNICA DE AGUASCALIENTES

BACHELOR'S DEGREE IN ENGINEERING TECHNOLOGIES IN
INFORMATION TECHNOLOGY AND DIGITAL INNOVATION



Sprint #1 Deliverable 2

KAVEM System

COURSE

Integrative Project II

Period

April - August 2026

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Introduction

The following content presents the results and advances of Sprint #1. In this sprint, we focused on the inventory table, products, and materials management. At the same time, we worked on the design and began developing the program with a simple UX/UI but with basic functionality like adding a product and deleting a product. The project was managed using Jira.

Functional and Non-Functional Requirements

Functional Requirements (FR)

- **FR01 - Real-Time Inventory Management:** The system must allow real-time stock control displayed by size and product type.
- **FR02 - Production Module and Status:** The system must link production status with inventory availability to provide realistic delivery times.
- **FR03 - Product and Price Catalogue (CRUD):** The system must allow users to create, read, update, and delete (CRUD) products with their respective sizes and base prices (Life Jackets: Flea, Infant, Junior, Medium, Adult, Extra Large; Swimming Boards and Sliders).
- **FR04 - Quotation Automation:** The system must automatically calculate final prices, including VAT (16%), optional freight based on the delivery zone, and apply discounts if applicable.
- **FR05 - Document Generation (PDF):** The system must automatically export professional quotations in PDF format in less than 2 minutes.
- **FR06 - Customer Management:** The system must allow users to register customers and store their purchase history to facilitate recurring sales.
- **FR07 - Low Stock Alerts:** The system must generate automatic visual notifications when the inventory of any product runs low.
- **FR08 - Dashboard and Reports:** The system must provide a reporting interface to visualize the general status of the business.

Non-Functional Requirements (NFR)

- **NFR01 - Interface Usability (UX/UI):** The system must provide a simple and intuitive web-based interface optimized for warehouse and sales employees, incorporating KAVEM's corporate identity (company logo and color palette).
- **NFR02 - Response Time Efficiency:** The generation and processing of a complete quotation must not exceed 2 minutes.
- **NFR03 - Synchronization and Performance:** The processing and update time of inventory data must be reflected within a maximum of 15 minutes (ideally in real time to mitigate double-entry errors).
- **NFR04 - Architecture:** The system must be a web-based environment application.
- **NFR05 - Scalability:** The system design must lay the technical foundation for future electronic invoicing integration.

Constraints and Base Prices

Official Base Prices (VAT and Freight Not Included):

- **Life Jackets:** Flea (\$120), Infant (\$135), Junior (\$150), Medium (\$165), Adult (\$180), Extra Large (\$220)
- **Swimming Boards:** One Size (\$100)
- **Sliders:** Small (\$120), Junior (\$190), Medium (\$250), Adult (\$300)

Tax: The mandatory applicable VAT is 16% on the base price.

Use Case Diagram

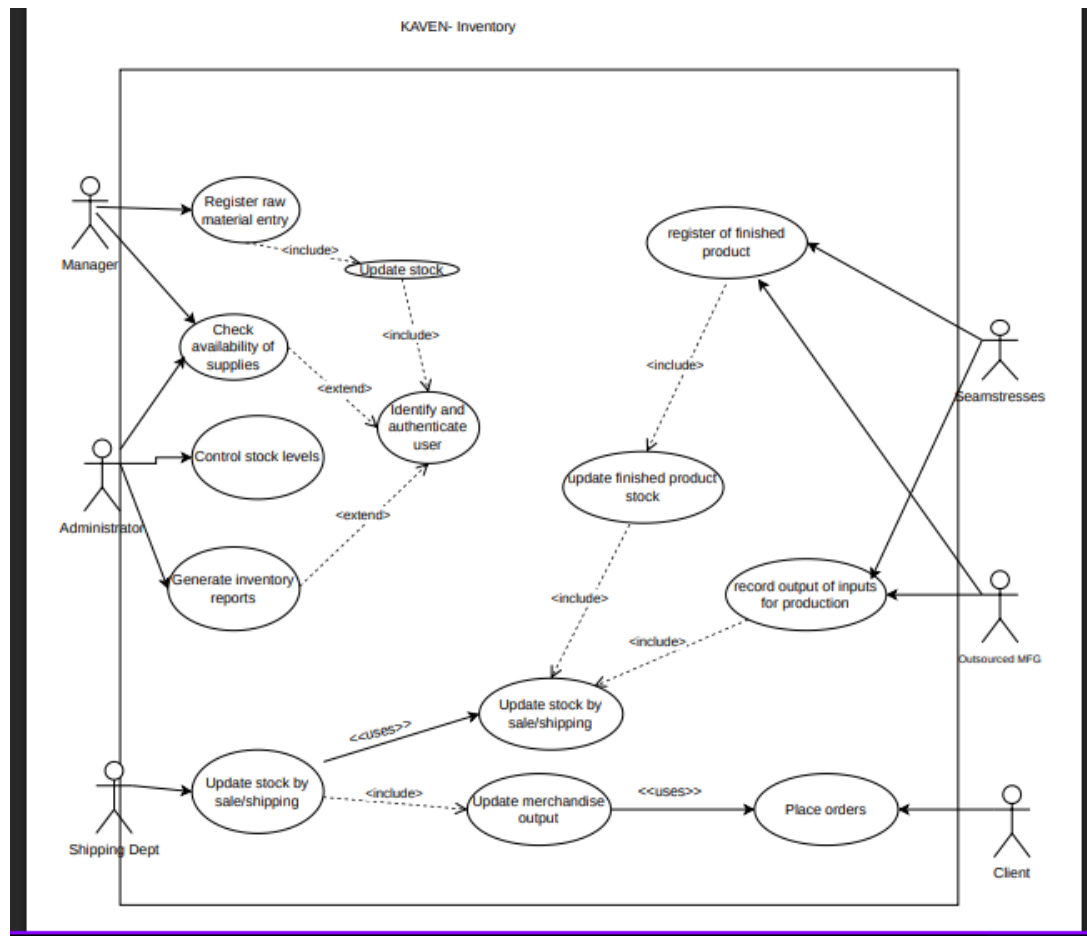


Figure 1: Use Case Diagram for KAVEM System

Main Actors:

- Warehouse Manager
- Seamstresses / Production Staff
- Administration
- Shipping Department
- Client

Main Use Cases:

- Input Entry Record
- Material Output for Production
- Minimum Stock Alert
- Manage Inventory

- Generate Quotation
- Place Orders

Use Case Scenario Cards

UC-01: Input Entry Record

Element	Description
Scenario	The Manager receives a shipment of foam and fabric for vests
Main Actor	Warehouse Manager
Precondition	The supplier has delivered the material and there is a prior purchase order
Main Flow	1. The Manager accesses the "Raw Materials" module 2. Scan or type the input code 3. Enter the amount received 4. The system adds the quantity to the current stock
Result	Raw material inventory is updated and Management can see availability

UC-02: Material Output for Production

Element	Description
Scenario	A Seamstress requests material to make a batch of 50 vests
Main Actor	Seamstresses
Precondition	There must be sufficient stock of fabric and buckles in the warehouse
Main Flow	1. The Seamstress requests the supplies from the system 2. The system checks if the requested quantity is available 3. Exit is authorized and a digital voucher is generated 4. The system automatically discounts meters of fabric from inventory
Result	The stock of raw materials is reduced and monitoring of "Work in Process" begins

UC-03: Minimum Stock Alert

Element	Description
Scenario	The system detects that there are few finished surfboards left after a mass shipment
Main Actor	Administration
Precondition	A "Reorder Point" (minimum allowed) has been previously defined for tables
Main Flow	1. The Shipping area records the departure of the last order 2. The system calculates that the current stock is less than the configured minimum 3. The system sends an automatic notification to the Administration module 4. Administration generates a production order for Maquila
Result	Shortages are avoided and new production of boards is planned

Database Entity-Relationship Diagram

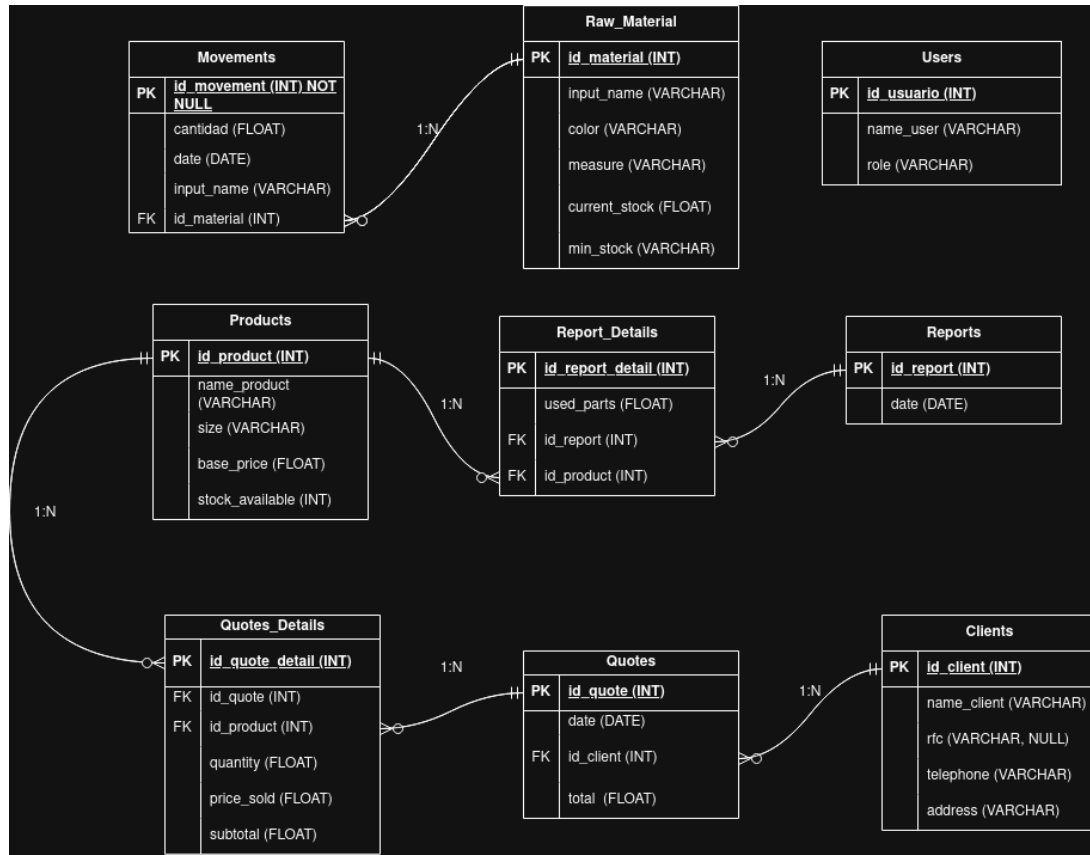


Figure 2: Entity-Relationship Diagram for Sprint #1

Main Entities

- **Product:** id, name, type, base_price, created_at
- **Inventory:** id, product_id, size, quantity, last_updated
- **Raw Material:** id, name, unit, quantity, min_stock
- **Customer:** id, name, email, phone, address
- **Quotation:** id, customer_id, date, subtotal, vat, total
- **Production Order:** id, product_id, quantity, status, date

Interface Design (Mockups)

Color Palette

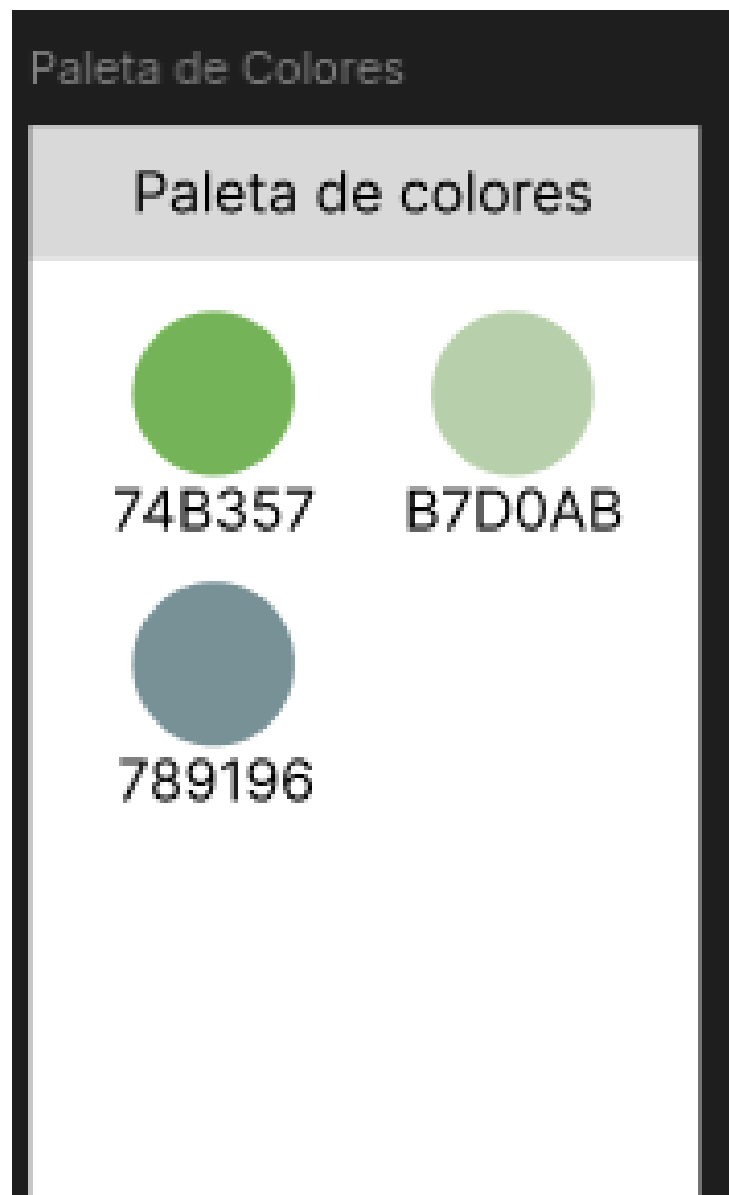


Figure 3: KAVEM Color Palette: #74B357, #789196, #B7D0AB

Main Menu Mockup



Figure 4: Main Menu - Inventory, Products, Materials

Products Inventory Mockup



Figure 5: Products Inventory Screen

Materials Mockup



Figure 6: Materials List Screen

Product Details Mockup (Life Jacket)



Figure 7: Product Details - Life Jacket with Size and Color Selection

Navigation Design

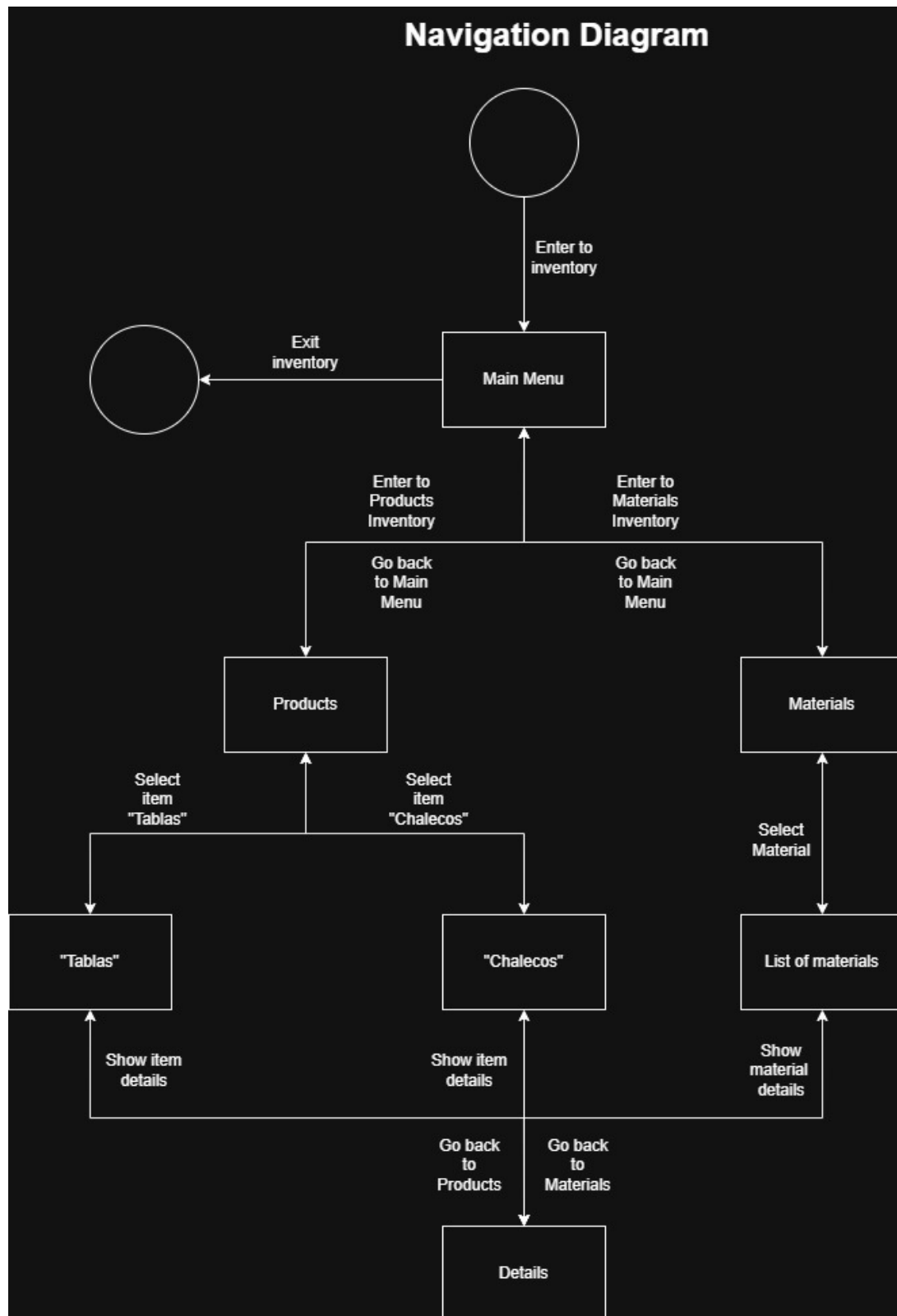


Figure 8: Navigation Flow Diagram for KAVEM System

Navigation Structure

- **Main Menu**

- Enter to Inventory
 - * Products Inventory
 - Select item "Tablas" (Swimming Boards)
 - Show item details
 - Go back to Products
 - * Materials
 - List of materials
 - Select material
 - Show material details
 - Go back to Materials
 - * Go back to Main Menu
- Exit Inventory

Test Cases

Sprint #1 Focus: Inventory Management basic operations (Create, Read, Delete)

Test Cases for Product Management

Test Case ID	TC-INV-01
Module	Inventory Management
Test Title	Verify adding a new product to inventory
Priority	High
Preconditions	User is authenticated and on the Inventory screen
Test Data	Product Name: "Chaleco Estándar", Type: "Life Jacket", Size: "Medium", Base Price: \$165, Quantity: 10
Execution Steps	1. Click "Add Product" button 2. Fill in product name: "Chaleco Estándar" 3. Select Type: "Life Jacket" 4. Select Size: "Medium" 5. Enter Base Price: 165 6. Enter Quantity: 10 7. Click "Save"
Expected Results	New product appears in the inventory list with all correct data
Actual Results	When you add a product, the system shows the product added
Status	Pass
Tester	Olivia Lizbeth Chairez Alba
Date	May 17, 2026

Test Case ID	TC-INV-02
Module	Inventory Management
Test Title	Verify deleting a product from inventory
Priority	High
Preconditions	At least one product exists in the inventory list
Test Data	Product to delete: "Chaleco Estándar"
Execution Steps	1. Locate product "Chaleco Estándar" in inventory list 2. Click the "Delete" button next to the product 3. Confirm deletion in the popup dialog

Expected Results	Product is removed from the inventory list and no longer appears
Actual Results	The product was deleted and no longer appears
Status	Pass
Tester	Olivia Lizbeth Chairez Alba
Date	May 17, 2026

Test Case ID	TC-INV-03
Module	Inventory Management
Test Title	Verify retrieving/displaying all products from inventory
Priority	High
Preconditions	At least one product exists in the database
Test Data	None
Execution Steps	1. Navigate to the Inventory module 2. Observe the displayed product list
Expected Results	System displays all products with their correct information (name, type, size, price, quantity)
Actual Results	The system shows all the product information
Status	Pass
Tester	Olivia Lizbeth Chairez Alba
Date	May 17, 2026

Test Case ID	TC-INV-04
Module	Inventory Management
Test Title	Verify adding a product with empty required fields (validation)
Priority	Medium
Preconditions	User is on the "Add Product" form
Test Data	Product Name: [leave blank], Type: "Life Jacket", Size: "Medium", Price: 165, Quantity: 10
Execution Steps	1. Leave "Product Name" field empty 2. Fill all other fields correctly 3. Click "Save"
Expected Results	System shows validation error: "Product name is required" and does NOT save the product

Actual Results	Re-	The system does not show a message when a required field is left empty
Status		Don't Pass
Tester		Olivia Lizbeth Chairez Alba
Date		May 17, 2026

Test Case ID	TC-INV-05
Module	Inventory Management
Test Title	Verify adding a product with negative quantity (validation)
Priority	Medium
Preconditions	User is on the "Add Product" form
Test Data	Product Name: "Chaleco Test", Type: "Life Jacket", Size: "Adult", Base Price: 180, Quantity: -5
Execution Steps	1. Fill all fields correctly 2. Enter negative quantity: -5 3. Click "Save"
Expected Results	System shows validation error: "Quantity must be zero or positive"
Actual Results	The system does not show an alert message when a negative value is entered
Status	Don't Pass
Tester	Olivia Lizbeth Chairez Alba
Date	May 17, 2026

Test Case ID	TC-INV-06
Module	Inventory Management
Test Title	Verify deleting a product that doesn't exist
Priority	Low
Preconditions	User is on Inventory screen
Test Data	Product ID that does not exist in database
Execution Steps	1. Attempt to delete a product with non-existent ID via API/direct call
Expected Results	System returns error message: "Product not found"
Actual Results	The system does not show an alert message when the product does not exist; it only throws an error in the console

Status	Don't Pass
Tester	Olivia Lizbeth Chairez Alba
Date	May 17, 2026

Test Summary for Sprint #1

Test Case ID	Description	Status	Implemented
TC-INV-01	Add new product	Pass	Yes
TC-INV-02	Delete product	Pass	Yes
TC-INV-03	Get/display all products	Pass	Yes
TC-INV-04	Validation - empty fields	Don't Pass	No
TC-INV-05	Validation - negative quantity	Don't Pass	No
TC-INV-06	Delete non-existent product	Don't Pass	No

Table 7: Test Cases completed in Sprint #1

Screenshots of the Program in Operation

Screenshot 1: Main Screen

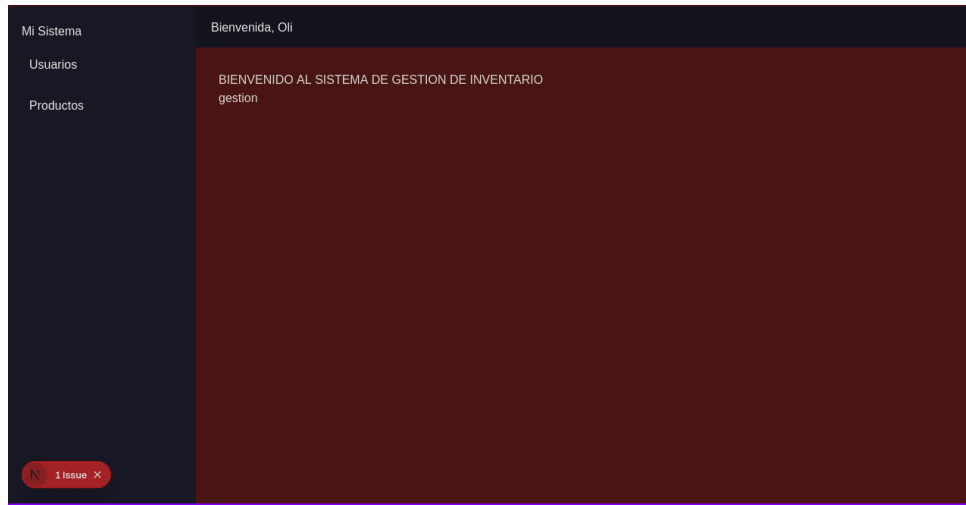


Figure 9: Screen that shows a welcome message

Screenshot 2: Users manager

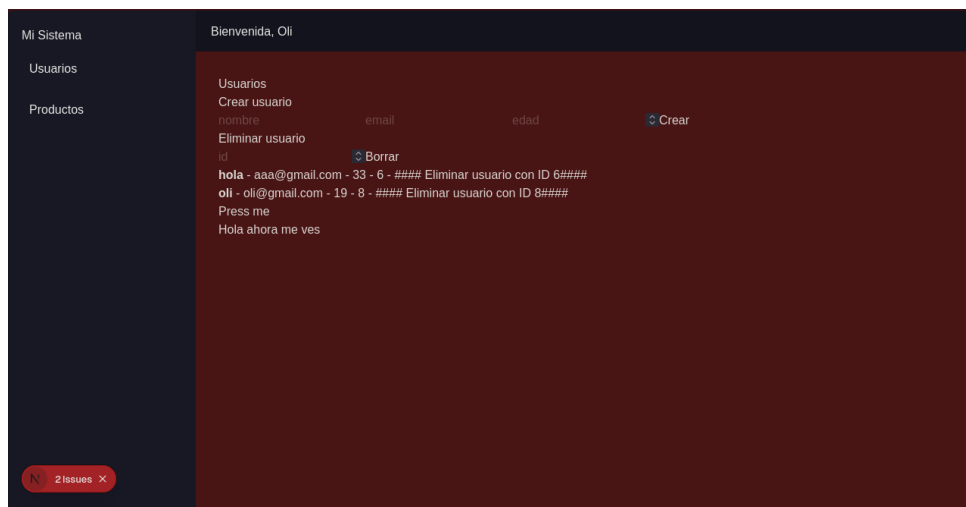


Figure 10: Users manager screen showing some users

Screenshot 3: Products Inventory

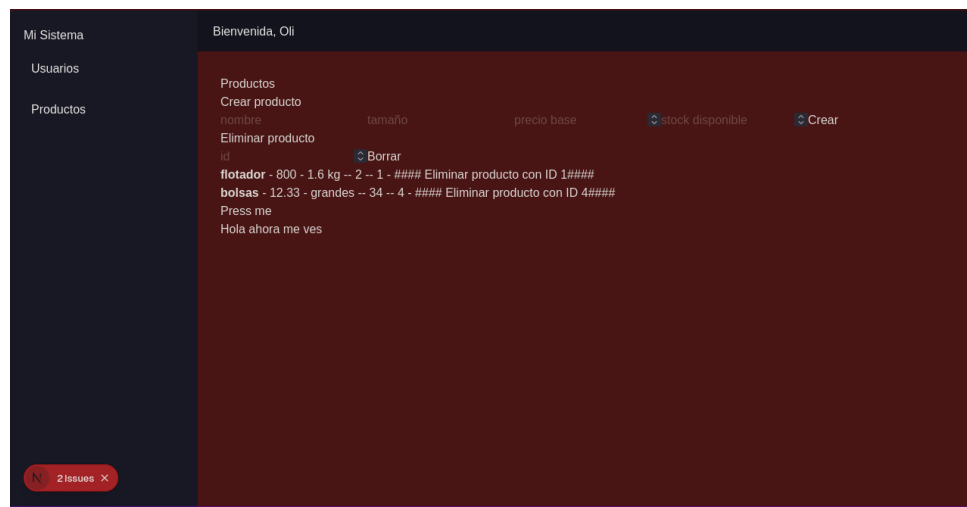


Figure 11: Products inventory screen showing life jackets, swimming boards, and sliders

Sprint Management Using Jira

Sprint #1 Overview

Sprint Duration	May 14 – May 20, 2026
Total Tasks	8
Completed Tasks	8
Completion Percentage	100%

Jira Board Screenshot

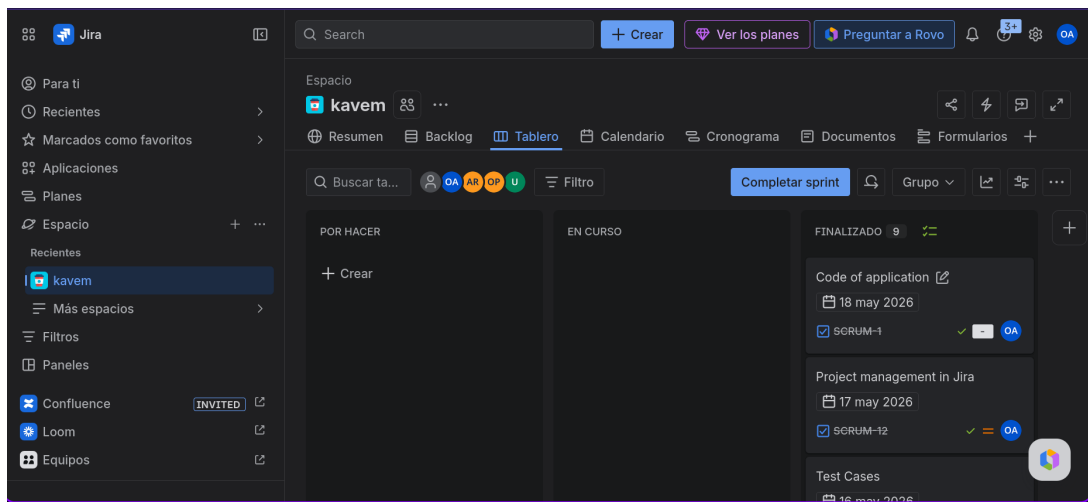


Figure 12: Jira Sprint Board for Sprint #1

Tasks Completed in Sprint #1

Issue ID	Task Name	Assigned To	Status
SCRUM-6	Scenario Cards (One for each Use Case)	Omar Facio Palos	Finalizado
SCRUM-7	Entity-Relationship Diagram	Noé Ezequiel Estrada Ramírez	Finalizado
SCRUM-8	Functional & Non-Functional Requirements	Noé Ezequiel Estrada Ramírez	Finalizado
SCRUM-9	Navigation Diagram	Alan Ricardo Jauregui De La Riva	Finalizado
SCRUM-10	Test Cases	Olivia Lizbeth Chairez Alba / Alan Ricardo Jauregui De La Riva	Finalizado

SCRUM-11	User Interface Design (Mock-ups)	Alan Ricardo Jauregui De La Riva	Finalizado
SCRUM-12	Use Case Diagram	Omar Facio Palos	Finalizado
SCRUM-13	Code of the Application	Olivia Lizbeth Chairez Alba	Finalizado

Table 8: Tasks completed during Sprint #1

Task Assignments Summary

- **Olivia Lizbeth Chairez Alba (UP240594)** - Code development (SCRUM-13), Test Cases (SCRUM-10)
- **Noé Ezequiel Estrada Ramírez (UP240770)** - ER Diagram (SCRUM-7), Requirements (SCRUM-8)
- **Omar Facio Palos (UP240470)** - Use Case Diagram (SCRUM-12), Scenario Cards (SCRUM-6)
- **Alan Ricardo Jauregui De La Riva (UP240960)** - Navigation Diagram (SCRUM-9), Mockups (SCRUM-11), Test Cases (SCRUM-10)

Burndown Chart

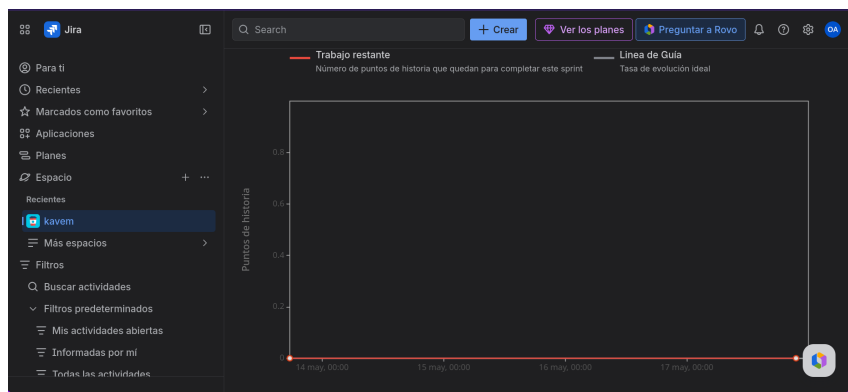


Figure 13: Sprint #1 Burndown Chart from Jira